

Bhargav Srinivasan

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RESEARCH INTERESTS

I am interested in machine learning methods and theory for genomics, focusing on modeling and mechanistic interpretation of structural, epigenetic, and regulatory variation.

EDUCATION

University of Maryland

B.S. in Computer Science & B.S. in Biological Sciences

- Computer Science + Entomology Departmental Honors, Gemstone Honors Program

College Park, MD

Expected May 2027

GPA: 3.8/4

SELECT RESEARCH EXPERIENCES

Complex SV Identification — Kolmogorov Lab, National Cancer Institute

May 2026 – Present

Student Intern

Bethesda, MD

- Developing a deep learning method to identify and localize complex structural variation in cancer genomes.

Raw Signal Analysis Methods — Firtina Lab, UMD

Oct 2025 – Present

Student Research Assistant

College Park, MD

- Building methods for nanopore sequencing raw signal analysis: methylation detection, error correction, assembly.

Molecular Surveillance — AstraZeneca, Vaccines & Immune Therapies

May 2025 – Aug 2025

Bioinformatics Intern

Gaithersburg, MD

- Engineered and deployed a bacterial genome-processing pipeline supporting vaccine R&D surveillance efforts.

- Developed an unsupervised learning approach to identify non-typeable *Haemophilus influenzae* cell-type clusters.

Insect Genome Bioinformatics Pipeline — Fritz Lab, UMD

Feb 2024 – Present

Computational Biology Research Assistant

College Park, MD

- Analyzing WGS short-read data to determine basis of pesticide resistance through copy number variation.

- Benchmarking short-read structural variation and copy number variation callers in collaboration with Molloy Lab.

Fish Behavior Classification — Gemstone Program, UMD

Aug 2023 – Present

Student Research Assistant, Mentor Liaison

College Park, MD

- Developing a computer vision aided classification model to identify the social behaviors of Lake Malawi Cichlids.

HONORS & AWARDS

1. ISMB HiTSeq Best Poster Award

RawMod: A Deep Learning Approach to Modification Agnostic Classification in Nanopore Sequencing

ISMB: Intelligent Systems for Molecular Biology, Washington, D.C., USA. July 2026.

2. Recipient of the Honors College Research Grant, CS Professional Development Fund, GO Fund, and Ernest N. Cory Undergraduate Scholarship

Funds collectively totaling over \$5,000 in support of independent research and conference travel.

SELECT PUBLICATIONS

1. Simon Ambrozak, Ulysse McConnell, Bhargav Srinivasan, Burak Ozkan, Ernest Zhang, and Can Firtina, “CRANE: Correcting Errors in Raw Nanopore Signals Using Hidden Markov Models”, [arXiv](#), 2026. [\[link\]](#)

2. Bhargav Srinivasan, Archana Venkataraman, and Srinivasa N. Raja, “Artificial Intelligence and Pain Management: Cautiously Optimistic”, [Editorial] [Pain Management](#), Sep. 2024. [\[link\]](#)

3. Bhargav Srinivasan, Simon Ambrozak, and Can Firtina, “RawMod: A Deep Learning Approach to Modification Agnostic Classification in Nanopore Sequencing”, **in preparation**.

SELECT PRESENTATIONS

1. **RawMod: A Deep Learning Approach to Modification Agnostic Classification in Nanopore Sequencing**
Poster Presentation, [ISMB: Intelligent Systems for Molecular Biology](#), Washington, D.C., USA. July 2026.

2. **A Deep Learning Approach to Identifying Complex Structural Variation in the Genome**
Invited Seminar Talk, [Cancer Data Science Laboratory](#), National Cancer Institute, Bethesda, MD, USA. July 2026.
Poster Presentation & Flash Talk, [NCI Summer School on Algorithmic Cancer Biology](#), Bethesda, MD, USA. July 2026.

3. **A Machine Learning Approach to Identify Unique Non-typeable *Haemophilus influenzae* (NTHi) Clusters Using Lipooligosaccharide (LOS) Biosynthetic Genes**
Poster Presentation, [ASM Microbe 2026](#), Washington, D.C., USA. June 2026.

4. **Correlations between Read Depth, Copy Number Variation, and Bt Resistance in *Helicoverpa Zea***
Poster Presentation, [PAG33: Plant and Animal Genome Conference](#), San Diego, CA, USA. January 2026.